ISLAND *Immunity*

Investigating Jeju Farm Animal Diseases

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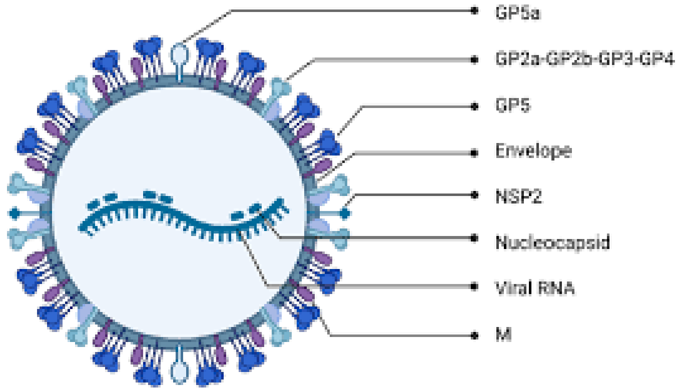
- Pathology Profiles: Swine, Bovine, Equine, & Avian
- Epidemiological Insights: Tracking Infection Trends
- Biosecurity: The Island's First Line of Defense

NLCS JEJU Veterinary Society

Porcine Reproductive and Respiratory Syndrome Virus (PRRSV)

What is PRRSV?

PRRSV is an enveloped, single-stranded RNA virus belonging to the family *Arteriviridae*. It primarily targets pigs and is one of the most economically significant pathogens in global swine production. The virus exists in two major genotypes: the European leystad virus (Type 1) and the North American VR-2332 (Type 2), both of which can cause reproductive failure in sows and severe respiratory disease in pigs. PRRSV is highly mutable, which complicates vaccine development.

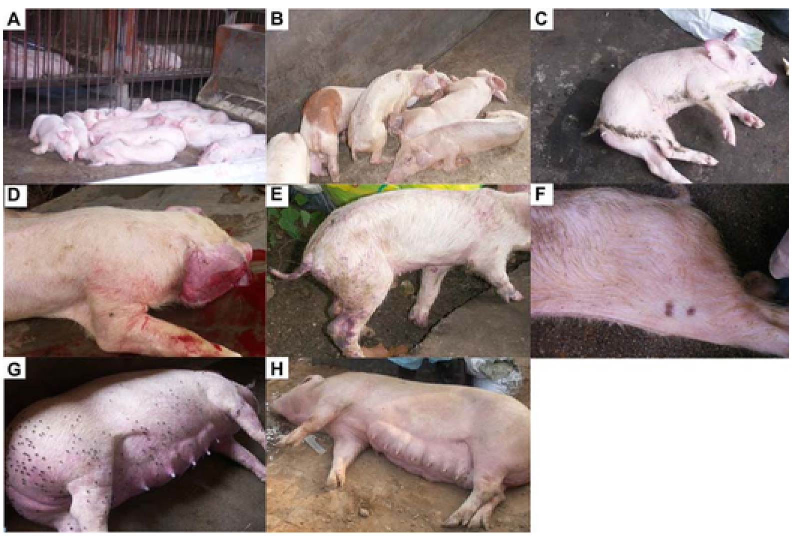


Why this matters to Jeju

Jeju is famous for its Black Pork (Heuk-dwaeji), a cultural and economic icon of the island. Virus like PRRSV affect farm revenue and threaten a key part of Jeju's heritage. Because PRRSV spreads easily through the air, maintaining strict biosecurity on Jeju's island farms is critical to protecting our unique local breed.

Clincal Symptoms

- Adult pigs: Reproductive failure including abortions, stillbirths, weak-born piglets, and delayed return to estrus
- Piglets (2–8 weeks old): Fever, labored breathing, poor weight gain, and increased susceptibility to secondary infections due to immunosuppression
- Piglets display lethargy and wasting (A-C). Characteristic skin discoloration and cyanosis (blueing) of the ears and abdomen, often referred to as "Blue Ear Disease" (D-F, right image).



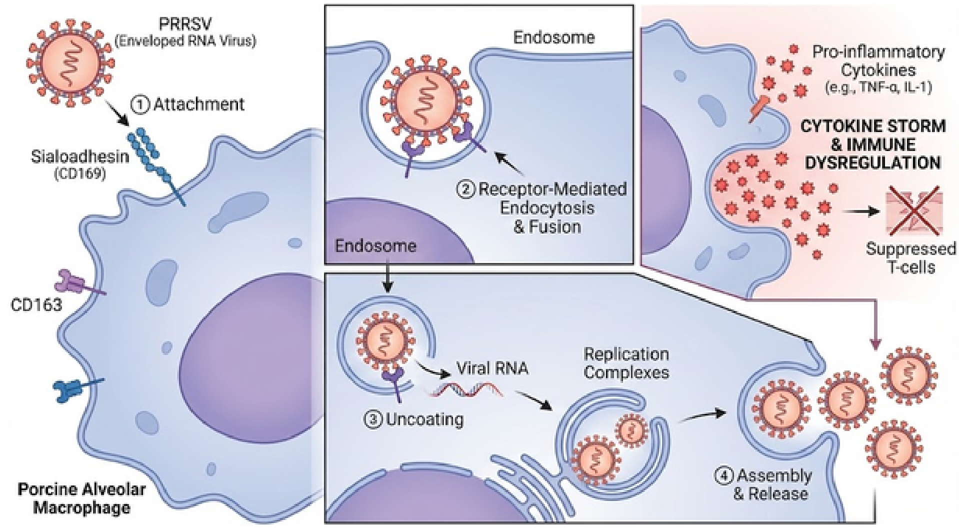
Mechanism of infection

As shown in the diagram, the infection follows a lethal pathway:

① The virus primarily targets alveolar macrophages in the lungs. Instead of destroying the virus, these immune cells become the host.

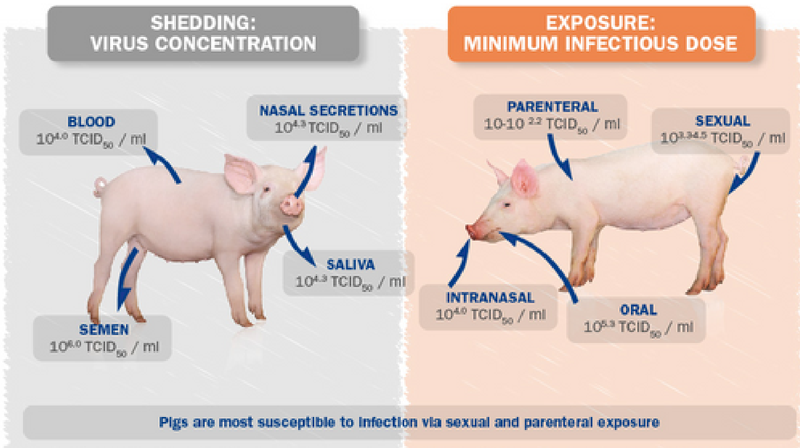
② As seen in the receptor close-up, the virus enters via receptor-mediated endocytosis. It exploits specific surface molecules—CD163 and sialoadhesin—using them like a key to unlock the cell and slip inside.

③ Once inside, PRRSV disrupts the pig's immune system. The diagram highlights the result: a release of inflammatory signals known as a Cytokine Storm, leading to tissue damage and vascular failure.



Transmission pathways

- **Direct contact**: Nose-to-nose contact, sexual intercourse
- **Aerosolized droplets**: PRRSV can spread in the air under certain conditions, enabling farm-to-farm transmission
- **Contaminated fomites and feed**: Equipment, clothing, vehicles, and feed can carry infectious virus particles
- **Vertical transmission**: Pregnant sows can transmit the virus to fetuses, leading to reproductive failure



Epidemology in Jeju

Data from the Jeju Veterinary Research Institute (2023) shows that PRRSV infection rates increased by 18% from 2021 to 2023, with peaks aligning with periods of high feed import activity. According to the Korea Animal Health Integrated System (KAHIS), 111 pigs on reported farms were infected with PRRSV in 2025 by October. These data highlight the seasonal and anthropogenic factors contributing to viral spread.

Control Strategies

- **Biosecurity**: Maintaining hygiene, controlling access to farms, and disinfecting vehicles and equipment reduce the risk of viral introduction.
- **Vaccination**: Both modified-live and inactivated vaccines help prevent PRRSV infection, though high viral mutation rates can limit their efficacy.
- **Herd Immunity**: Exposing pigs to vaccines via shared fodder helps protect the wider pig population.
- **AI Monitoring:** Early detection through AI-driven monitoring systems, such as sensors, cameras, and GPS, can identify outbreaks before widespread transmission. Monitoring DNA-protein information and predicting suitable vaccines also helps prevent the spread of the virus.
- **Farm Management**: Limiting animal movement and quarantining new or sick animals reduces the risk of farm-to-farm transmission. Timely reporting to KAHIS is also essential.

Conclusion

PRRSV remains a persistent challenge for Jeju pig farming due to its high mutation rate, multiple transmission pathways, and immunosuppressive effects. Prevention strategies combining biosecurity, vaccination, and herd management are critical to mitigating its impact on animal health and economic productivity.

Bone Viral Diarrhea Virus (BVDV)

What is it

Bovine Viral Diarrhea Virus (BVDV) is a highly contagious viral disease affecting cattle worldwide. Despite its name, it impacts far more than the digestive system—BVDV weakens immunity, disrupts reproduction, reduces growth, and increases susceptibility to other infections. The virus occurs in two main genotypes (BVDV-1 and BVDV-2) and can cause either mild or severe illness depending on the strain and the animal's immune status.

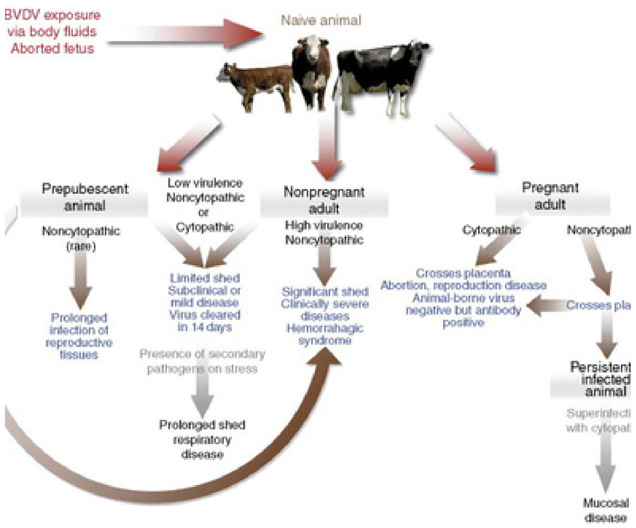
Clincal Symptoms

Mild:

- fever
- reduced appetite
- nasal discharge
- drops in milk yield

Severe:

- diarrhea
- dehydration
- mouth ulcer
- embryonic death



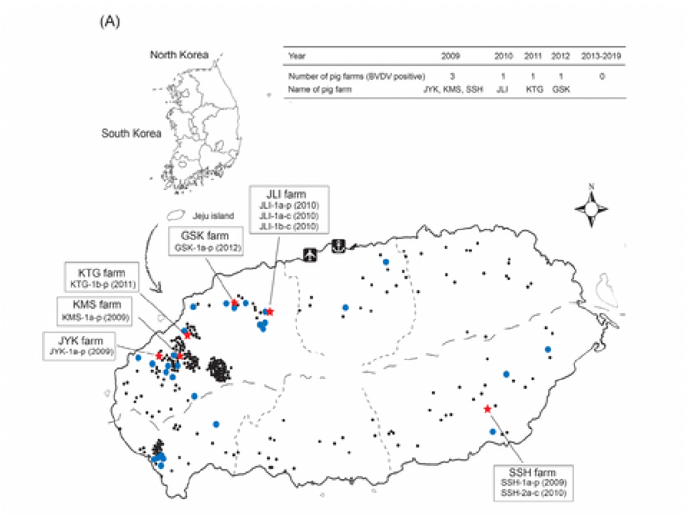
Transmission pathways

The virus spreads through:
Direct contact: nose-to-nose contact, shared water troughs, or contaminated equipment.
PI calves: the most significant and persistent source of infection.
Infected pregnant cows: transmitting the virus across the placenta.
Animal movement: purchasing untested animals introduces BVDV into clean herds.
Fomites: boots, clothing, needles, or vehicles that carry contaminated secretions.

Mechanism of infection

BVDV primarily targets immune cells, especially those in lymphoid tissues. Once inside the body, the virus suppresses the animal's immune system, making it vulnerable to secondary diseases like pneumonia. If the fetus is infected during early gestation, its immune system mistakes BVDV as "self," resulting in PI calves that continuously shed high levels of virus. These PI animals are the engine of long-term BVDV circulation.

Epidemology in Jeju



Jeju Island's mixed cattle–pig farming environment and frequent farm-to-farm movement create ideal conditions for BVDV persistence. Studies on Jeju farms have shown continuous virus circulation, often traced back to undetected PI animals. The island's relatively dense cattle population—especially Hanwoo beef herds—means that a single PI calf can spread infection rapidly if routine testing is not implemented.

Economic impacts

BVDV reduces overall herd productivity by lowering fertility, decreasing milk yield, slowing growth rates, and increasing calf mortality. These biological losses translate into higher expenditures for veterinary care and replacement animals, placing disproportionate pressure on small and mid-scale farms. At the industry level, persistent infection undermines regional production efficiency and disrupts trade by creating variability in herd health status. Public agencies must respond through coordinated monitoring systems and regulatory oversight, as unmanaged BVDV circulation complicates livestock planning across the island. This makes the disease not only a veterinary concern but also a factor influencing long-term agricultural policy and resource allocation on Jeju.

Control Strategies

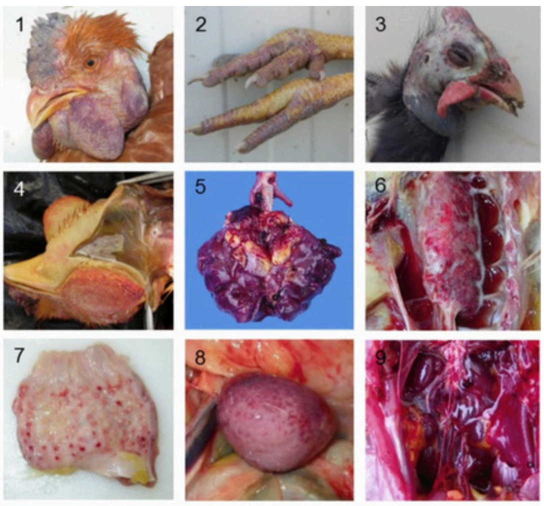
- PI animal detection and removal: Routine antigen/PCR testing of newborn calves and high-risk groups to identify persistently infected individuals.
- Pre-movement screening: Testing animals before sale or transfer to prevent introducing undetected infections into clean herds.
- Vaccination: Use of inactivated or modified-live vaccines—especially before breeding—to protect fetuses from transplacental infection.
- Enhanced biosecurity: Regular disinfection of equipment, controlled visitor access, dedicated clothing/boots, and separation of age groups.
- Accurate record-keeping: Documenting test results, vaccination schedules, and animal movements to support long-term disease monitoring.

Avian Influenza

Causes

There are a lot of different causes but the main reasons are contact with infected poultry and contaminated equipment. Infected poultry are birds who are affected by disease, and within contact, the disease can spread to the chicken rapidly. This can also happen when the chickens have a direct contact contaminated with the poultry products. For example, when a chicken directly contacts with contaminated manure, litter, equipment, clothing, feed, or water, then the chicken who touched the items can be infected by the disease, which leads to the infection of the Avian influenza.

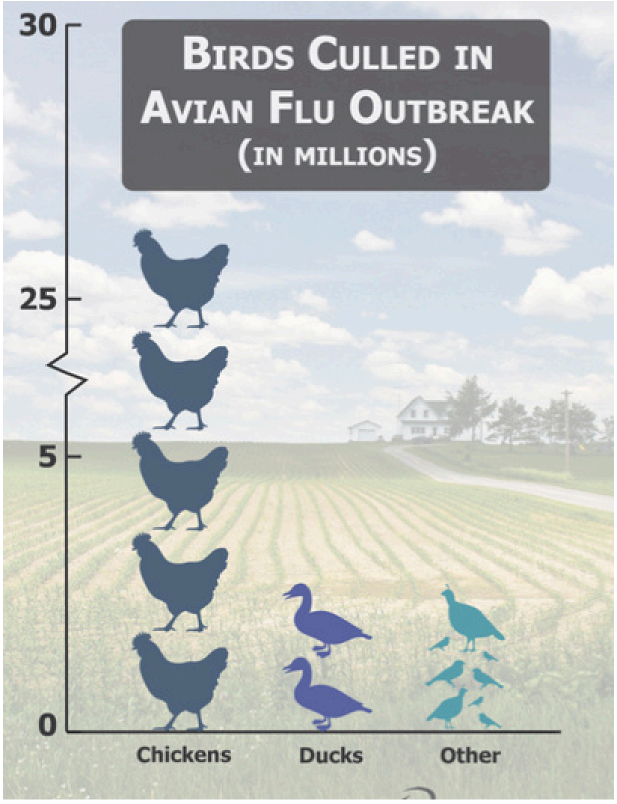
Clincal Symptoms



When chickens gets infected by Avian influenza, there are different symptoms that are shown throughout their body, which are like decreased energy and appetite, having nasal discharge(runny nose), ruffled feathers, purple discoloration throughout the comb, wattles, and legs, having diarrhea, Incoordination or lack of coordination, and more. Sometimes, a high number of chickens may die suddenly, often with no prior signs or any symptoms. For example, if you see picture 2, you can see how the leg, which is called shank, is starting to lose color and starting to turn purple. As well as in section b, you can see the comb, the fleshy red-colored outgrowth, is also losing color with the size becoming smaller.

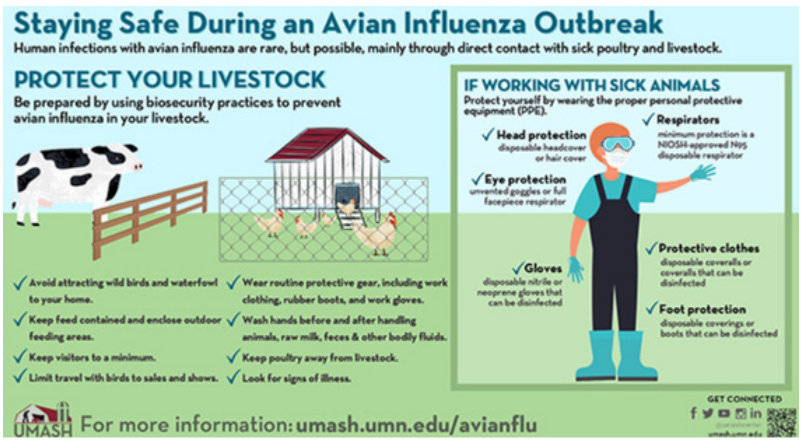
Outbreak and Impact

In January 2015, June 2017, and the recent of December 2023 and February 2024, there was an outbreak of avian influenza in Jeju island which caused a lot of damage toward the environment, as well as huge financial losses. In 2015, about 120,000 chickens and ducks near 21 poultry farms on Jeju Island were culled after they were confirmed to be infected with a highly pathogenic H5N8(Avian influenza) strain. There was a record 37.8 million birds were culled from last November to that month, incurring 1 trillion won ($892 billion) in losses. There were also other outbreaks of avian influenza that occurred in South Korea at various times, including a major nationwide outbreak from late 2016 to April 2017 and another in late 2020, the specific incident involving chickens on Jeju Island was centered around June 2017.



Control Strategies

Therefore, it is crucial to take action when you spot a chicken who has Avian influenza. As an individual, we need to Report Immediately, Isolate the Sick Bird, and Implement Strict Biosecurity. The individual should contact a local veterinarian, or contact the national/state animal health authority to help the infected chickens. They should also isolate the sick chicken to prevent further infection toward other chickens. During the process, the individual should not touch the sick or dead birds, and should not touch contaminated surfaces without wearing personal protective equipment since it can also infect humans as well. Therefore after quickly removing the infected chickens, they should clean and disinfect all coops, equipment, feeders, and waterers using an EPA-approved disinfectant.



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As a government reaction, they should start doing an active monitoring on the poultry farms, bird markets, and slaughterhouses. Additionally, doing mandatory reporting toward farm owners, veterinarians, and the public about the dead or sick birds to stop the disease from spreading furthermore. The government should also provide financial support and additional compensation to farmers for stock losses and rapid response to ensure cooperation.

For future development, the government should do a superior policy development of refining and improving existing emergency response plans and develop more advanced vaccines toward the livestock as well.

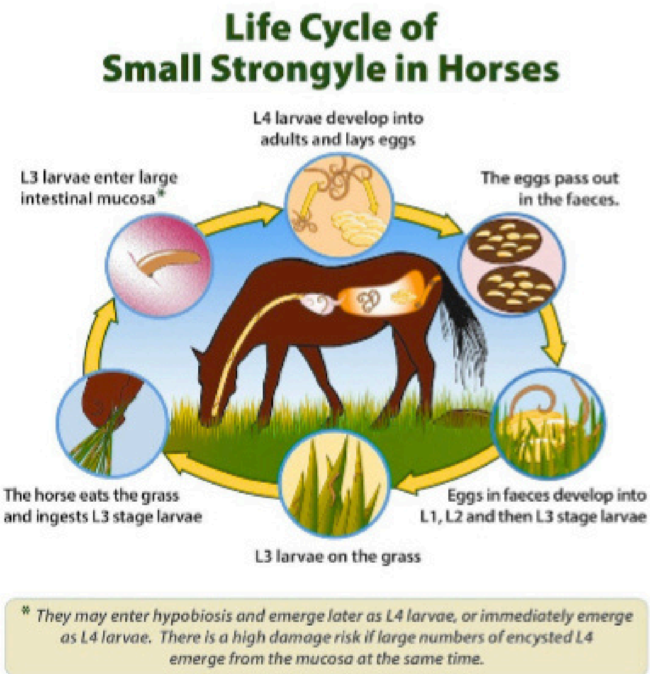
Therefore, understanding avian influenza and having proper reaction to protect animal health, supporting farmers, and ensuring safer and more resilient poultry systems in the future is going to be an essential source to reduce the cause of Avian influenza toward the chickens.

Strongylus vulgaris

Strongylus vulgaris (The Common Strongyle)

Phylum: Nematoda | Class: Rhabditia | Order: Strongylida | Superfamily: Strongyloidea | Family: Strongylidae

The common strongyle (Strongylus spp.) belonging to the subfamily Strongylinae of the family Strongylidae; is a parasite primarily inhabiting the large intestine of horses. S. vulgaris is one of three Strongylus spp that infect horses. The equine bloodworm Strongylus vulgaris is regarded as the most pathogenic equine GI helminth. Historically, the prevalence of S. vulgaris was reported to be 80–100%. Two species infect horses: Parascaris equorum and Parascaris univalens. Ascarid eggs are environmentally durable and remain infectious for several years, though they do not tolerate temperatures > 40°C. Eggs likely remain viable through winter in colder climates, becoming infective in the following foaling season.



Life Cycle and Migration

third-stage larvae (L3) exsheath in the small intestine, invade the mucosa of the small intestine, cecum, and colon within 2–3 days, and migrate subendothelially in intestinal arterioles. After 3 weeks, they reach the cranial mesenteric artery and its major branches. There, they molt to the fourth stage (L4) and enter the arterial lumen, later molting to the L5 stage. Eventually, they are carried down the circulatory tree to form abscesses (approx. 1 cm) in the walls of the cecum and colon. These abscesses rupture, releasing larvae into the intestinal lumen to mature. The complete life cycle takes approximately 6 months, with 4 months spent in the mesenteric arteries. In temperate climates, abundance is seasonal; horses acquire infections during grazing, migration occurs in winter, and egg-shedding adults peak in the spring.

Pathology and Clinical Signs

Larvae migrating along the arterial intima damage the endothelium, simultaneously forming and residing within thrombi. While cranial mesenteric artery aneurysms occur, the parasite generally causes thrombotic endarteritis and panarteritis. In the large intestine mucosa, larvae can cause hemorrhagic necrotic enteritis due to parasitic granulomatous embolism and diffuse eosinophilic infiltration. Many horses remain asymptomatic, though some suffer intestinal dysfunction, notably colic. Mechanisms may include compromised intestinal blood flow and interference with autonomic ganglia near the cranial mesenteric artery root. Thrombotic arteritis is hypothesized as a common colic cause, with concurrent hindlimb spasms and recurring colic reported in foals .
Experimental work suggests larvae newly arrived in the intestinal wall (pre-migration) may also significantly cause colic. Notably, many horses with arterial lesions never experience colic, and vice versa. Recent opinion favors cyathostomins over S. vulgaris as a primary cause of equine colic.

treatment and warning

The current recommendation is to include an anthelmintic in the treatment plan, but this carries risks. Treating ascarid-associated colic with an anthelmintic can exacerbate intestinal impaction; therefore, this approach should be chosen only when pain is manageable and gastric reflux is absent.

- **Benzimidazoles**: Generally recommended due to low resistance rates and a nonparalytic mode of action (killing worms slowly over 2–3 days). A full labeled dose should be administered with gastric reflux checks every 3 hours for 24–48 hours.
- **Pyrantel**: Effective but used with caution; its paralytic mode of action could theoretically lead to impaction.
- **Macrocyclic Lactones**: Unlikely to work well due to high rates of resistance worldwide.

Progression should be monitored with frequent ultrasonographic examination. If anthelmintic treatment is chosen, dead worms can be recovered via nasogastric tube or left to pass in feces. Supportive measures (fluids, pain meds, antimicrobials) should be implemented as appropriate.

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Prevention

To minimize disease risk, current recommendations include fecal diagnostic testing, routine monitoring of efficacy, and targeted treatments. For ascarids, it is generally recommended to target Parascaris spp twice before weaning (at approximately 2 months and 5 months old). Benzimidazoles are the anthelmintic of choice for prevention. Fecal egg counts before and after the 5-month treatment help determine the relative presence of strongyles versus ascarids and evaluate treatment efficacy.